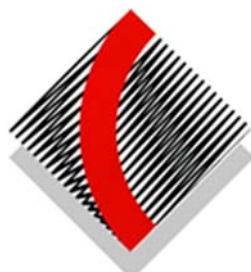


NAME: _____

CLASS: 11MTA___ or 11MTX___

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2019

YEAR 11

AP1

MATHEMATICS ADVANCED

Time allowed – 1.5 hours plus 5 minutes reading time

DIRECTIONS TO CANDIDATES:

- Write your name on the question/answer paper.
- Attempt all questions.
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- For all questions show relevant mathematical reasoning and/or calculations.
- NESA approved calculators may be used.

Topic	Total
Question 1	/19
Question 2	/9
Question 3	/29
Question 4	/13
Total	/70

70 marks

Attempt All Questions

Answer the questions in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the back of the examination paper.

Question 1	(19 marks)	Marks
------------	------------	-------

- | | | |
|-----|-----------------------------|---|
| (a) | Expand $(2x - 6y)(2x + 6y)$ | 1 |
|-----|-----------------------------|---|

- | | | |
|-----|--|---|
| (b) | Express in simplest form $(3\sqrt{2} - 2)^2$ | 2 |
|-----|--|---|

- | | | |
|-----|--|---|
| (c) | Simplify $\sqrt{12} + \sqrt{18} - \sqrt{48}$ | 2 |
|-----|--|---|

- | | | |
|-----|---|---|
| (d) | Solve $6 - 3x - 5x^2 = 0$ using the quadratic formula, correct to 3 decimal places. | 2 |
|-----|---|---|

- (e) Solve the quadratic equation $x^2 + 4x - 10 = 0$ by completing the square.
Leave your answer in surd form.

2

- (f) Simplify

(i) $\frac{x+3}{2} - \frac{2x-1}{5}$

2

(ii) $\frac{x^2 - 9}{x^2 - 2x - 15} \div \frac{x^2 - 5x + 6}{3x^2 - 12}$

3

- (g) Simplify $\frac{(3x^2)^2 \times \sqrt[3]{x}}{3x^5}$, leaving your answer without negative and fractional indices. **2**

- (h) Expand and simplify $\sqrt{2}(2\sqrt{10} + 9\sqrt{8}) - \frac{\sqrt{5}-2}{\sqrt{5}}$ with a rational denominator. **3**

End of Question 1

Question 2 (9 marks)

Marks

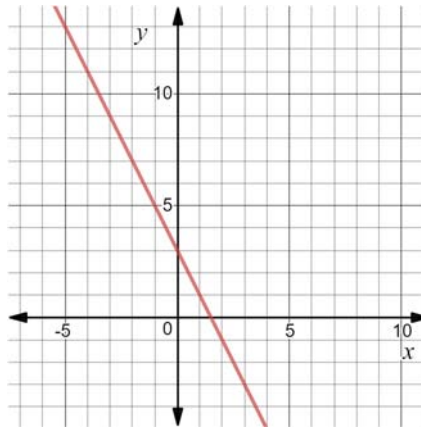
- (a) The function $f(x)$ is defined by $f(x) = \begin{cases} 4^x, & x \leq 1 \\ \frac{4}{x}, & x > 1 \end{cases}$. What is the value of $f(0 \cdot 5) + f(2)$?

- (A) 4
(B) 10
(C) 18
(D) 24

1

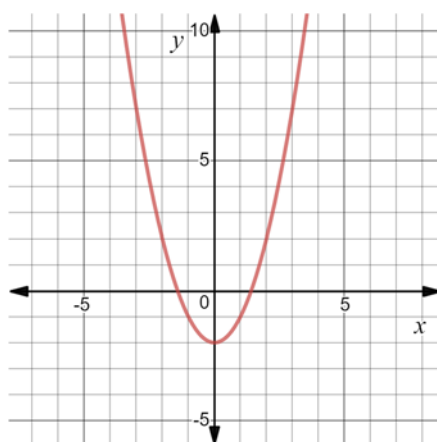
- (b) Determine **whether the following relations are also functions** and then categorise each graph as either one-to-one, one-to-many, many-to-one or many-to-many.

(i)



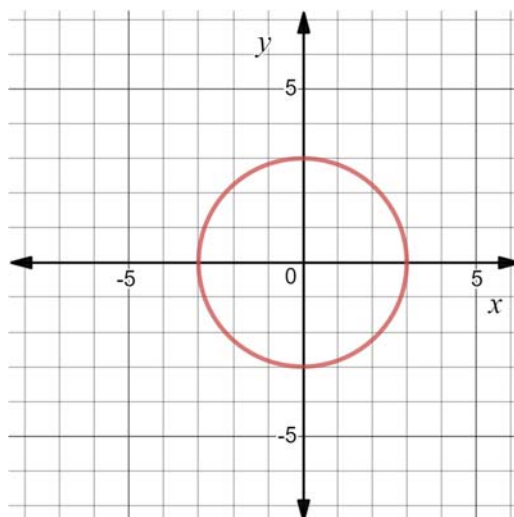
1

(ii)



1

(iii)



1

(c) Given that $f(x) = 3x^2 + 1$, find $f(-2)$

1

(d) If $A(x) = x^2 - 6$ and $B(x) = 11x - 2x^2$. Simplify

(i) $B(x) - A(x)$ **1**

(ii) $A(x)B(x)$ **1**

(iii) $A(B(x))$ **2**

End of Question 2

Question 3 (29 marks)**Marks**

(a) A line ℓ has equation $3y + 2x = 6$. What is the gradient of any line parallel to ℓ ?

(A) -2

(B) $-\frac{2}{3}$

(C) $\frac{3}{2}$

(D) 2

1

(b) What is the discriminant of $2x^2 + 3x + k = 0$?

(A) $\Delta = \sqrt{9 - 8k}$

(B) $\Delta = 9 - 8k$

(C) $\Delta = 9 - 12k$

(D) $\Delta = \sqrt{9 - 12k}$

1

(c) For $y = 2x^2 + 6x + 2$, the minimum value of y is:

(A) $-\frac{23}{2}$

(B) $-\frac{7}{2}$

(C) $-\frac{5}{2}$

(D) $\frac{31}{2}$

1

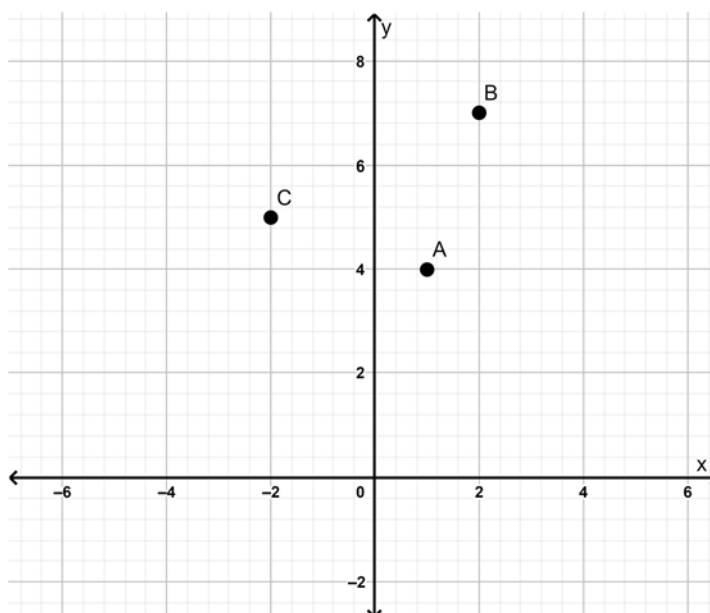
(d) Find the equation of the line in general form that passes through the points $(8,0)$ and $(0,6)$.

2

- (e) Find the values of p for which the equation $x^2 + (p - 1)x + 1 = 0$ has equal roots.

2

- (f) Three points A, B, C have coordinates $(1,4)$, $(2,7)$ and $(-2,5)$ respectively.



- (i) Calculate the length of BC , leave your answer in exact form.

1

- (ii) Show that the equation of the line BC is $x - 2y + 12 = 0$ 2

- (iii) Show that the equation of the line, ℓ , perpendicular to BC which passes through A is $2x + y - 6 = 0$. 2

- (iv) Hence, find the coordinates of P , the point where the ℓ meets the line BC . 2

- (g) For the following quadratic function $y = x^2 + 6x$.

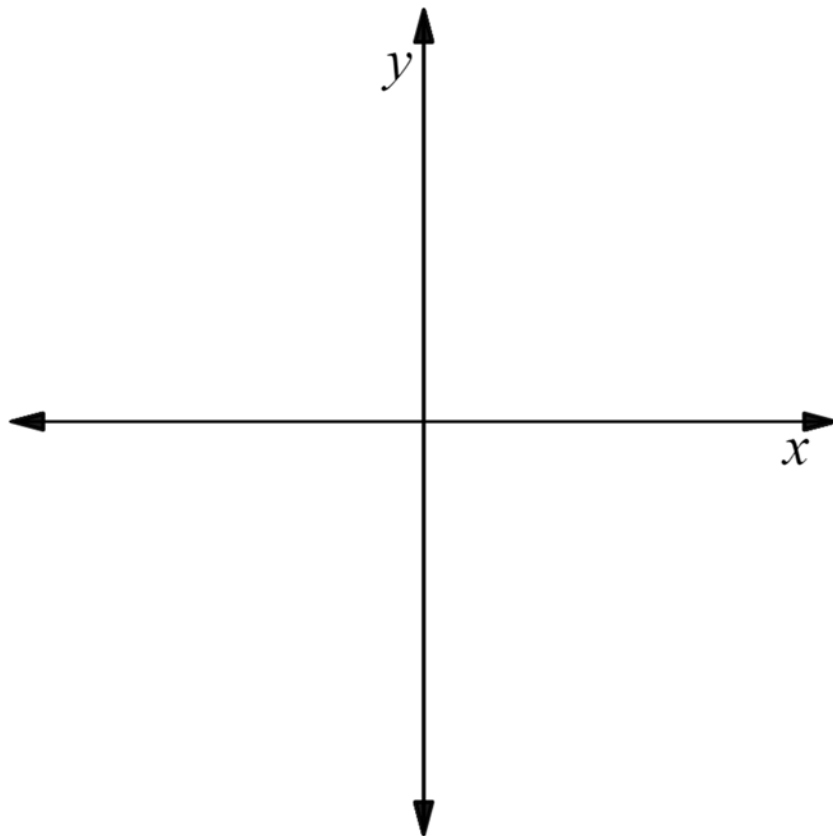
- (i) Find the x and y intercepts. 2

(ii) Find the coordinates of the vertex.

1

(iii) Neatly sketch the parabola $y = x^2 + 6x$, clearly showing all intercepts.

2



(iv) State the domain and range using interval notation.

2

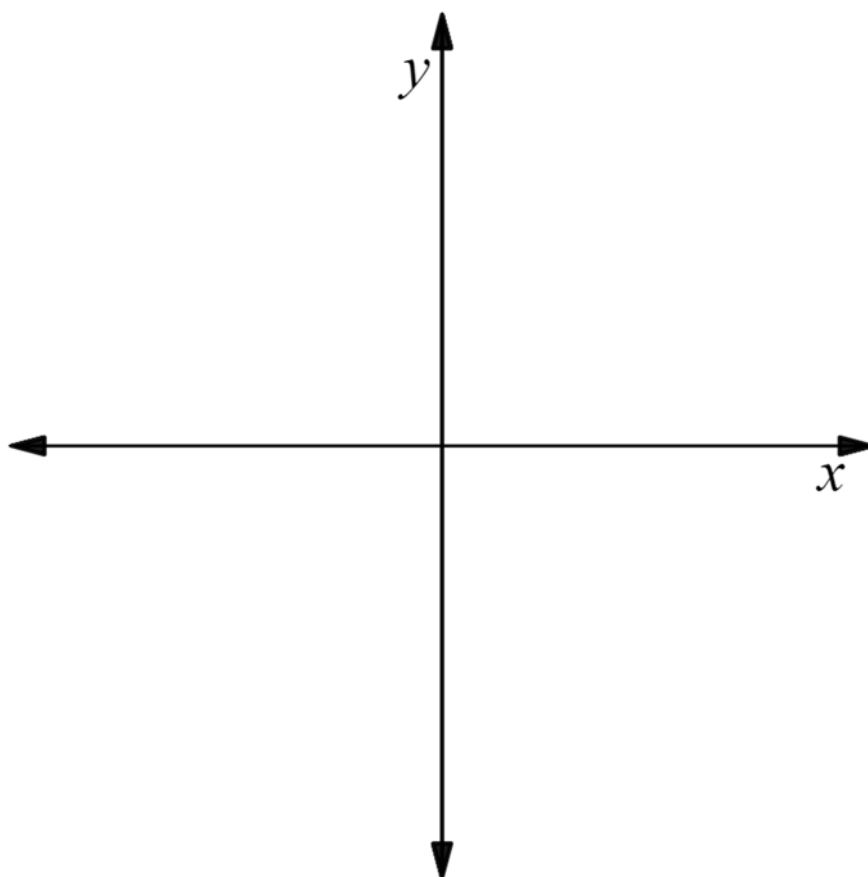
- (h) Solve the following simultaneous equations.

$$\begin{aligned}4x &= 12 + y \\ y &= x^2 - x - 12\end{aligned}$$

3

- (i) Sketch the graph of $y = -(x - 3)(x - 1)(x + 1)$, showing all essential features.

2



- (j) The cost of producing x litres of orange juice is $20 + x$ dollars. If the income from the sale of x litres of the juice is $2x^2$ dollars, calculate the number of litres that must be sold to make a profit of \$800. **3**

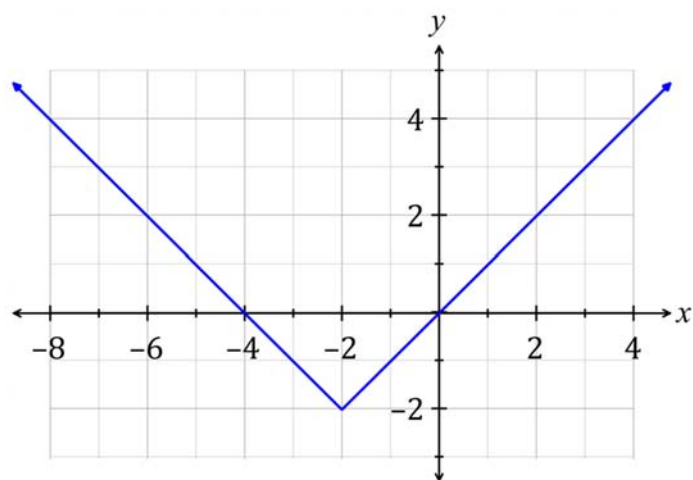
This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

End of Question 3

Question 4 (13 marks)

Marks

(a) What is the equation of the function below?



(A) $y = |x + 2| - 2$

(B) $y = |x - 2| + 2$

(C) $y = |2 - x| - 2$

(D) $y = |2 + x| + 2$

1

(b) For the polynomial $P(x) = 7x^6 + 5x^3 + 9x - 2$

(i) Find the degree of $P(x)$

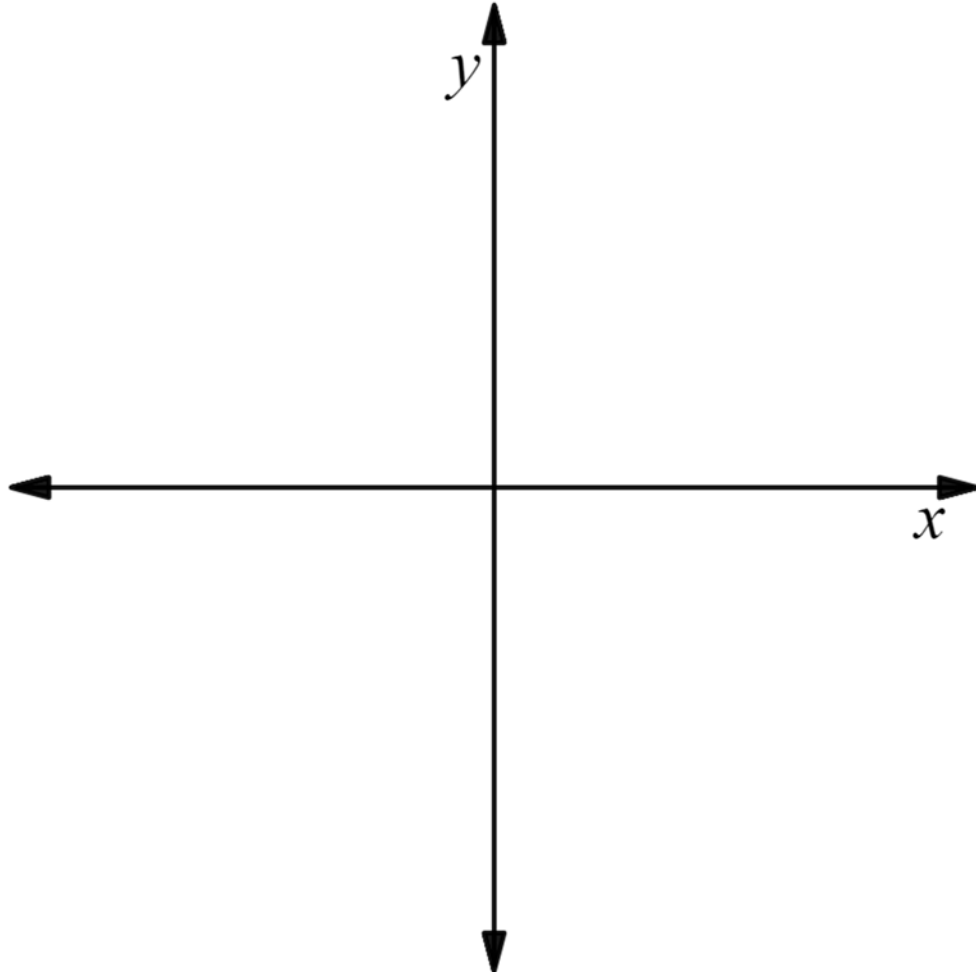
1

(ii) State the leading term

1

- (c) (i) Show that $x(x + 1)(x - 2)^2 = x^4 - 3x^3 + 4x$ 2

- (ii) Hence, sketch the graph of $P(x) = x^4 - 3x^3 + 4x$ 1



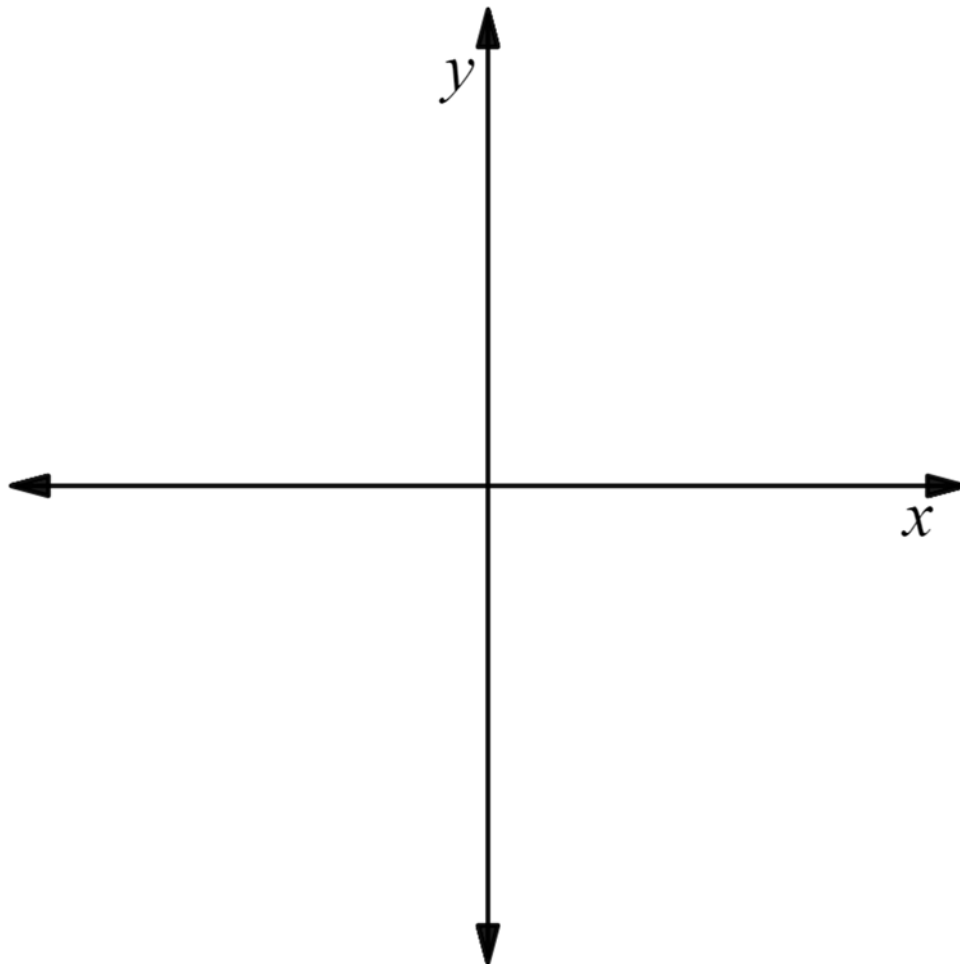
- (d) Consider the function $f(x) = \sqrt{9 - x^2}$
- (i) State the domain and range of $f(x)$, in interval notation. 2

- (ii) State whether the function $f(x)$, is odd, even or neither.
You must justify your answer algebraically.

2

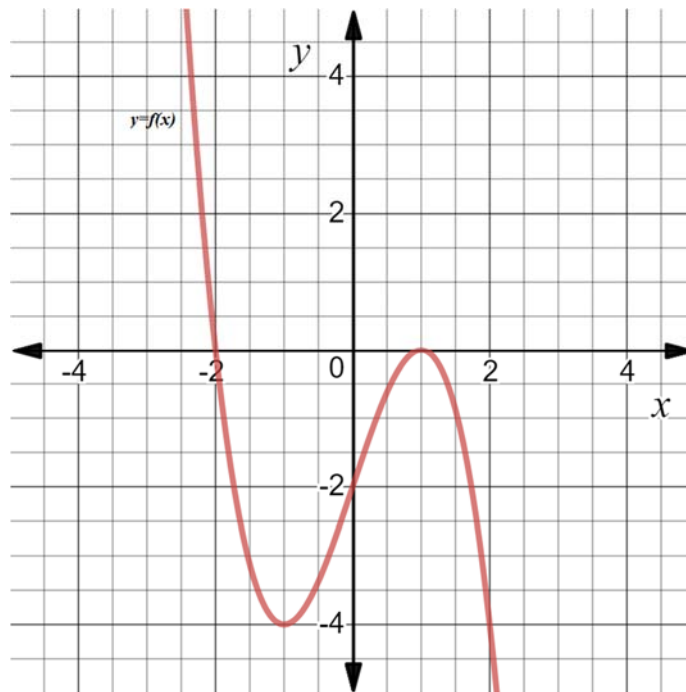
- (iii) Sketch the graph of $f(x) = \sqrt{9 - x^2}$, showing all intercepts.

1



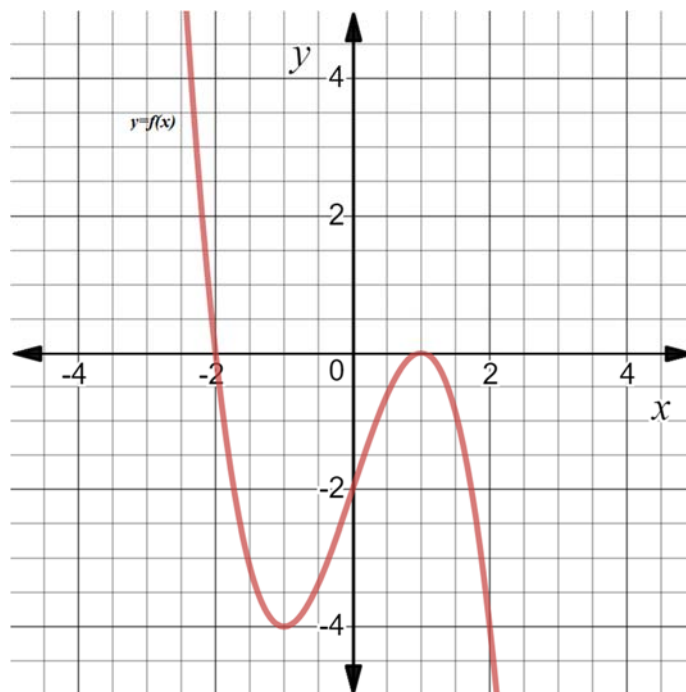
- (e) The graph of $y = f(x)$ is shown below. Sketch the graph of
- (i) $y = -f(x)$

1



- (ii) $y = -f(-x)$

1



End of Paper

Extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a full page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for writing. There are no margins, text, or other markings on the page.

[illegible]

70 marks

Attempt All Questions

Answer the questions in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the back of the examination paper.

Question 1 (19 marks)

Marks

- (a) Expand $(2x - 6y)(2x + 6y)$

1

$$4x^2 - 36y^2$$

- (b) Express in simplest form $(3\sqrt{2} - 2)^2$

2

$$\begin{aligned} &= (3\sqrt{2})^2 - 2 \times (3\sqrt{2}) \times 2 + 2^2 \\ &= 18 - 12\sqrt{2} + 4 \\ &= 22 - 12\sqrt{2} \end{aligned}$$

- (c) Simplify $\sqrt{12} + \sqrt{18} - \sqrt{48}$

2

$$\begin{aligned} &= 2\sqrt{3} + 3\sqrt{2} - 4\sqrt{3} \\ &= -2\sqrt{3} + 3\sqrt{2} \quad \text{or} \quad 3\sqrt{2} - 2\sqrt{3} \end{aligned}$$

- (d) Solve $6 - 3x - 5x^2 = 0$ using the quadratic formula, correct to 3 decimal places.

2

$$a = -5, b = -3, c = 6$$

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(-5) \times 6}}{2(-5)}$$

$$= \frac{3 \pm \sqrt{9 + 120}}{-10}$$

$$= \frac{3 \pm \sqrt{129}}{-10}$$

$$\therefore x = \frac{3 + \sqrt{129}}{-10} \quad \text{or} \quad x = \frac{3 - \sqrt{129}}{-10}$$

$$x = -1.436$$

$$x = 0.836$$

- (e) Solve the quadratic equation $x^2 + 4x - 10 = 0$ by completing the square.
Leave your answer in surd form.

2

$$x^2 + 4x = 10$$

$$x^2 + 4x + 4 = 10 + 4$$

$$(x + 2)^2 = 14$$

$$x + 2 = \pm \sqrt{14}$$

$$x = \pm \sqrt{14} - 2 \quad \text{or} \quad -2 \pm \sqrt{14}$$

- (f) Simplify

(i) $\frac{x+3}{2} - \frac{2x-1}{5}$

2

$$\frac{5(x+3)}{10} - \frac{2(2x-1)}{10}$$

$$= \frac{5x+15 - 4x+2}{10}$$

$$= \frac{x+17}{10}$$

(ii) $\frac{x^2 - 9}{x^2 - 2x - 15} \div \frac{x^2 - 5x + 6}{3x^2 - 12}$

3

$$= \frac{x^2 - 9}{x^2 - 2x - 15} \times \frac{3x^2 - 12}{x^2 - 5x + 6}$$

$$= \frac{(x-3)(x+3)}{(x-5)(x+3)} \times \frac{3(x+2)(x-2)}{(x-3)(x-2)}$$

$$= \frac{3(x+2)}{(x-5)} \quad \text{or} \quad \frac{3x+6}{x-5}$$

(g)

Simplify $\frac{(3x^2)^2 \times \sqrt[3]{x}}{3x^5}$, leaving your answer without negative and fractional indices.

2

$$= \frac{9x^4 \times x^{\frac{1}{3}}}{3x^5}$$

$$= 3x^4 \times x^{\frac{1}{3}} \times x^{-5}$$

$$= 3x^{4+\frac{1}{3}-5}$$

$$= 3x^{-\frac{2}{3}} \quad \text{or} \quad \frac{3}{x^{\frac{2}{3}}}$$

$$= \frac{3}{\sqrt[3]{x^2}} \quad \text{or} \quad \frac{3}{(\sqrt[3]{x})^2}$$

(h)

Expand and simplify $\sqrt{2}(2\sqrt{10} + 9\sqrt{8}) - \frac{\sqrt{5}-2}{\sqrt{5}}$ with a rational denominator.

3

$$= 2\sqrt{20} + 9\sqrt{16} - \frac{(\sqrt{5}-2) \times \sqrt{5}}{\sqrt{5} \times \sqrt{5}}$$

$$= 4\sqrt{5} + 36 - \frac{\sqrt{5}(\sqrt{5}-2)}{5}$$

$$= 4\sqrt{5} + 36 - \frac{(5-2\sqrt{5})}{5}$$

$$= \frac{20\sqrt{5} + 180 - 5 + 2\sqrt{5}}{5}$$

$$= \frac{22\sqrt{5} + 175}{5} \quad \text{or} \quad \frac{22\sqrt{5}}{5} + 35$$

End of Question 1

- (a) The function $f(x)$ is defined by $f(x) = \begin{cases} 4^x, & x \leq 1 \\ \frac{4}{x}, & x > 1 \end{cases}$. What is the value of $f(0.5) + f(2)$?

- (A) 4
(B) 10
(C) 18
(D) 24

$$f(0.5) = 4^{0.5} = 2$$

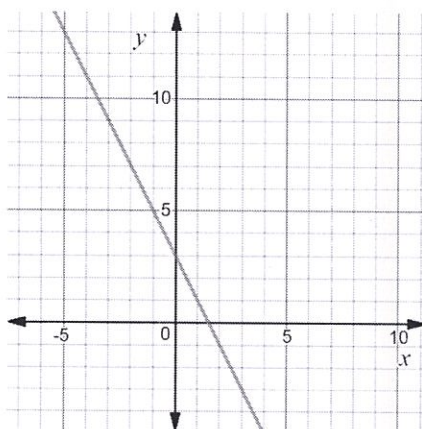
$$f(2) = \frac{4}{2} = 2$$

$$\therefore f(0.5) + f(2) = 2 + 2 = 4$$

1

- (b) Determine **whether the following relations are also functions** and then categorise each graph as either one-to-one, one-to-many, many-to-one or many-to-many.

(i)

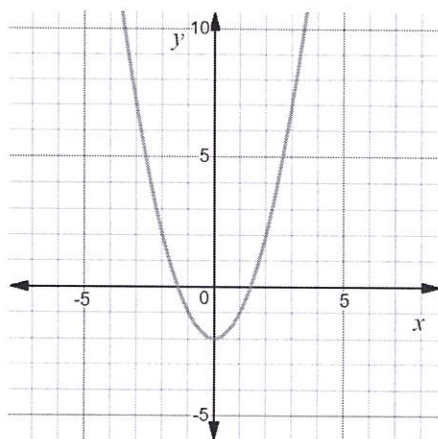


1

function

one to one

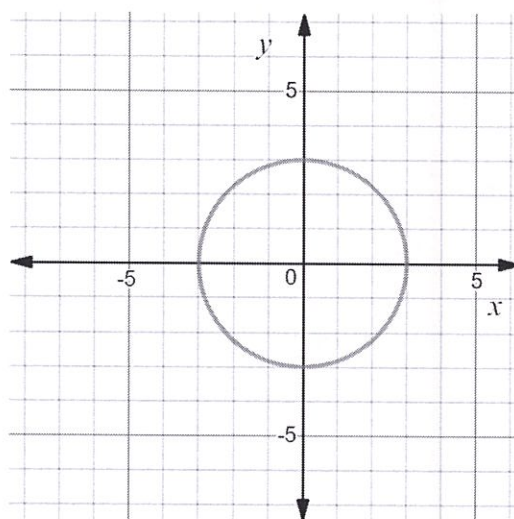
(ii)



1

function
many to one

(iii)



1

not a function - relation
many to many

(c) Given that $f(x) = 3x^2 + 1$, find $f(-2)$

1

$$\begin{aligned} f(-2) &= 3(-2)^2 + 1 \\ &= 12 + 1 \\ &= 13 \end{aligned}$$

(d) If $A(x) = x^2 - 6$ and $B(x) = 11x - 2x^2$. Simplify

(i) $B(x) - A(x)$

1

$$\begin{aligned} & 11x - 2x^2 - (x^2 - 6) \\ &= 11x - 2x^2 - x^2 + 6 \\ &= -3x^2 + 11x + 6 \end{aligned}$$

(ii) $A(x)B(x)$

1

$$\begin{aligned} & (x^2 - 6)(11x - 2x^2) \\ &= 11x^3 - 2x^4 - 66x + 12x^2 \end{aligned}$$

(iii) $A(B(x))$

2

$$\begin{aligned} & A(11x - 2x^2) \\ &= (11x - 2x^2)^2 - 6 \\ &= 121x^2 - 44x^3 + 4x^4 - 6 \\ &= 4x^4 - 44x^3 + 121x^2 - 6 \end{aligned}$$

End of Question 2

(a) A line ℓ has equation $3y + 2x = 6$. What is the gradient of any line parallel to ℓ ?

(A) -2

(B) $-\frac{2}{3}$

(C) $\frac{3}{2}$

(D) 2

$$y = -\frac{2}{3}x + 2$$

1

(b) What is the discriminant of $2x^2 + 3x + k = 0$?

(A) $\Delta = \sqrt{9 - 8k}$

(B) $\Delta = 9 - 8k$

(C) $\Delta = 9 - 12k$

(D) $\Delta = \sqrt{9 - 12k}$

$$\Delta = 9 - 8k$$

1

(c) For $y = 2x^2 + 6x + 2$, the minimum value of y is:

(A) $-\frac{23}{2}$

(B) $-\frac{7}{2}$

(C) $-\frac{5}{2}$

(D) $\frac{31}{2}$

$$x = \frac{-b}{2a}$$

$$= \frac{-6}{2 \times 2}$$

$$= -\frac{6}{4}$$

$$y = 2\left(-\frac{6}{4}\right)^2 + 6\left(-\frac{6}{4}\right) + 2$$

(d) Find the equation of the line in general form that passes through the points $(8,0)$ and $(0,6)$.

2

$$m = \frac{6-0}{0-8}$$

$$= \frac{6}{-8}$$

$$= -\frac{3}{4}$$

$$y - 0 = -\frac{3}{4}(x - 8)$$

$$4y = -3x + 24$$

$$3x + 4y - 24 = 0$$

- (e) Find the values of p for which the equation $x^2 + (p-1)x + 1 = 0$ has equal roots.

2

$$\Delta = 0$$

$$b^2 - 4ac = 0$$

$$(p-1)^2 - 4 \times 1 \times 1 = 0$$

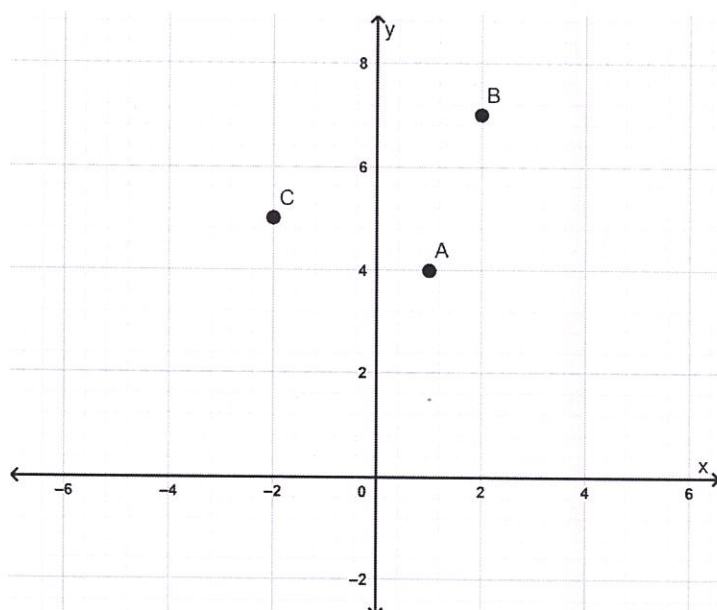
$$p^2 - 2p + 1 - 4 = 0$$

$$p^2 - 2p - 3 = 0$$

$$(p-3)(p+1) = 0$$

$$p = 3 \text{ or } p = -1$$

- (f) Three points A, B, C have coordinates $(1,4)$, $(2,7)$ and $(-2,5)$ respectively.



- (i) Calculate the length of BC , leave your answer in exact form.

1

$$B(2,7)$$

$$C(-2,5)$$

$$d = \sqrt{(2 - (-2))^2 + (7 - 5)^2}$$

$$= \sqrt{16 + 4}$$

$$= \sqrt{20}$$

$$= 2\sqrt{5}$$

- (ii) Show that the equation of the line BC is $x - 2y + 12 = 0$

2

$$m = \frac{7-5}{2-2}$$

$$= \frac{2}{0}$$

$$= \frac{2}{4}$$

$$= \frac{1}{2}$$

$$y - 7 = \frac{1}{2}(x - 2)$$

$$y - 7 = \frac{1}{2}x - 1$$

$$2y - 14 = x - 2$$

$$x - 2y + 12 = 0$$

- (iii) Show that the equation of the line, ℓ , perpendicular to BC which passes through A is $2x + y - 6 = 0$.

2

$$m_1 \times m_2 = -1$$

$$m = -2$$

$$y - 4 = -2(x - 1)$$

$$y - 4 = -2x + 2$$

$$2x + y - 6 = 0$$

- (iv) Hence, find the coordinates of P , the point where the ℓ meets the line BC .

2

$$x - 2y + 12 = 0 \quad \text{--- (1)}$$

$$2x + y - 6 = 0 \quad \text{--- (2)}$$

$$\text{--- } 2x - 4y + 24 = 0 \quad \text{--- (1a)}$$

$$5y - 30 = 0$$

$$5y = 30$$

$$\boxed{y = 6}$$

$$\text{sub } y = 6 \text{ into (1)}$$

$$x - 2(6) + 12 = 0$$

$$x - 12 + 12 = 0$$

$$\boxed{x = 0}$$

$$\therefore P(0, 6)$$

- (g) For the following quadratic function $y = x^2 + 6x$.

- (i) Find the x and y intercepts.

2

$$x\text{-intercept: let } y = 0$$

$$0 = x^2 + 6x$$

$$0 = x(x + 6)$$

$$x = 0 \text{ or } x = -6$$

$$y\text{-intercept: let } x = 0$$

$$y = 0^2 + 6(0)$$

$$y = 0$$

- (ii) Find the coordinates of the vertex.

1

$$y = x^2 + 6x$$

$$x\text{-coordinate } x = -3 \left(\frac{0 + -6}{2} \right)$$

sub $x = -3$ into eqn.

$$y = (-3)^2 + 6(-3)$$

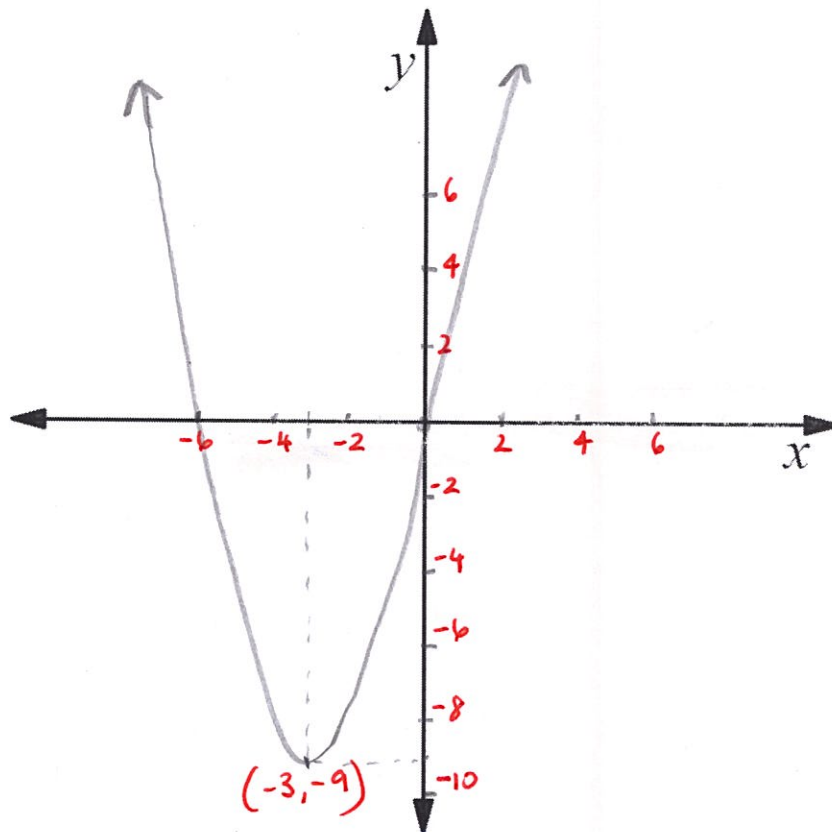
$$= 9 - 18$$

$$= -9$$

$$\therefore \text{Vertex } (-3, -9)$$

- (iii) Neatly sketch the parabola $y = x^2 + 6x$, clearly showing all intercepts.

2



- (iv) State the domain and range using interval notation.

2

$$\text{Domain: } (-\infty, \infty)$$

$$\text{Range: } [-9, \infty)$$

(h) Solve the following simultaneous equations.

$$4x = 12 + y \quad \text{--- ①}$$

$$y = x^2 - x - 12 \quad \text{--- ②}$$

Sub ② into ①

$$4x = 12 + (x^2 - x - 12)$$

$$4x = 12 + x^2 - x - 12$$

$$x^2 - 5x = 0$$

$$x(x - 5) = 0$$

$$x = 0 \text{ or } x = 5$$

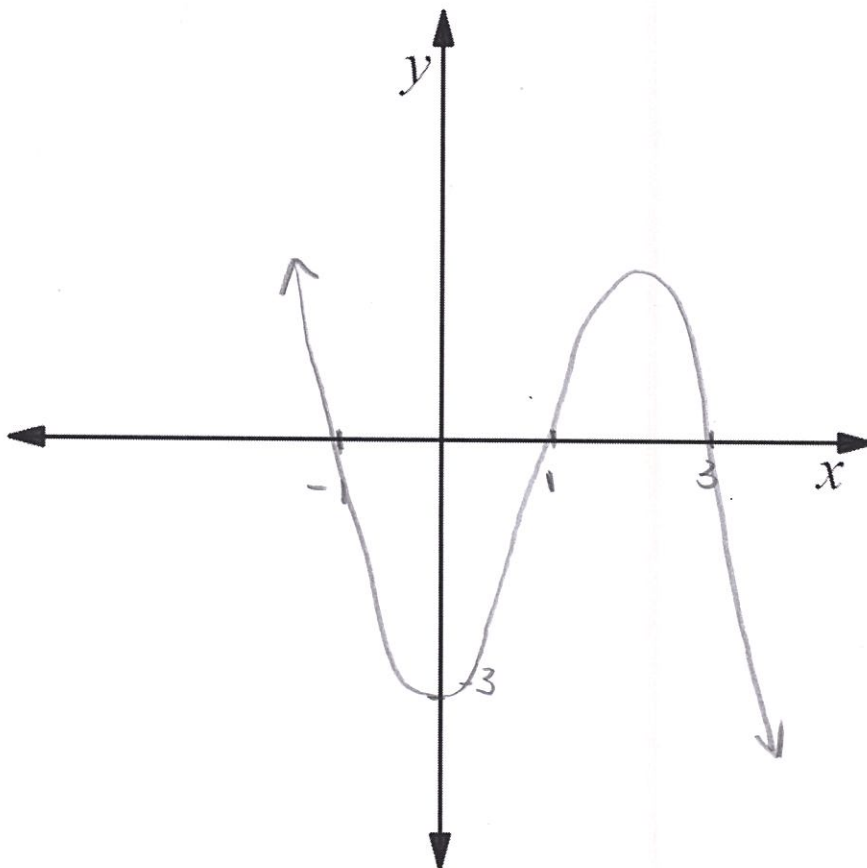
$$\text{When } x = 0, y = -12$$

$$\text{When } x = 5, y = 5^2 - 5 - 12$$

$$y = 8$$

$$\therefore (0, -12), (5, 8)$$

(i) Sketch the graph of $y = -(x - 3)(x - 1)(x + 1)$, showing all essential features.



- (j) The cost of producing x litres of orange juice is $20 + x$ dollars. If the income from the sale of x litres of the juice is $2x^2$ dollars, calculate the number of litres that must be sold to make a profit of \$800.

3

$$C = 20 + x$$

$$I = 2x^2$$

$$P = I - C$$

$$800 = 2x^2 - (20 + x)$$

$$800 = 2x^2 - 20 - x$$

$$2x^2 - x - 820 = 0$$

$$a = 2, b = -1, c = -820$$

$$x = \frac{-(-1) \pm \sqrt{1^2 - 4 \times 2 \times -820}}{2 \times 2}$$

$$= \frac{1 \pm \sqrt{1 + 6560}}{4}$$

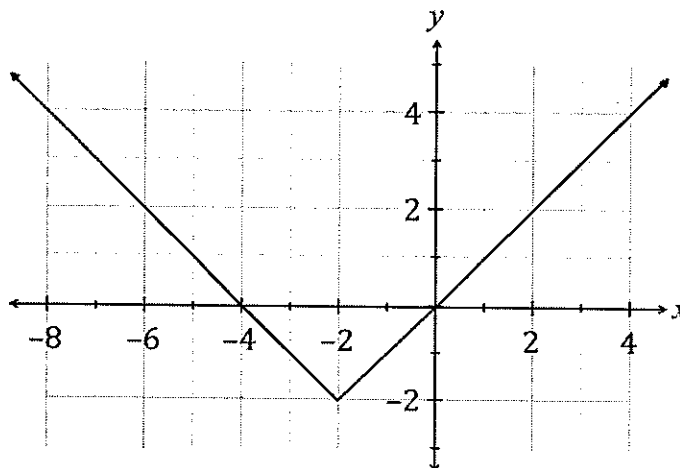
$$x = 20.5 \quad \text{or} \quad -20$$

$\therefore x$ cannot be negative.

$$\therefore x = 20.5 \text{ L.}$$

End of Question 3

(a) What is the equation of the function below?



(A) $y = |x + 2| - 2$

(B) $y = |x - 2| + 2$

(C) $y = |2 - x| - 2$

(D) $y = |2 + x| + 2$

1

(b) For the polynomial $P(x) = 7x^6 + 5x^3 + 9x - 2$

(i) Find the degree of $P(x)$

1

6

(ii) State the leading term

1

$7x^6$

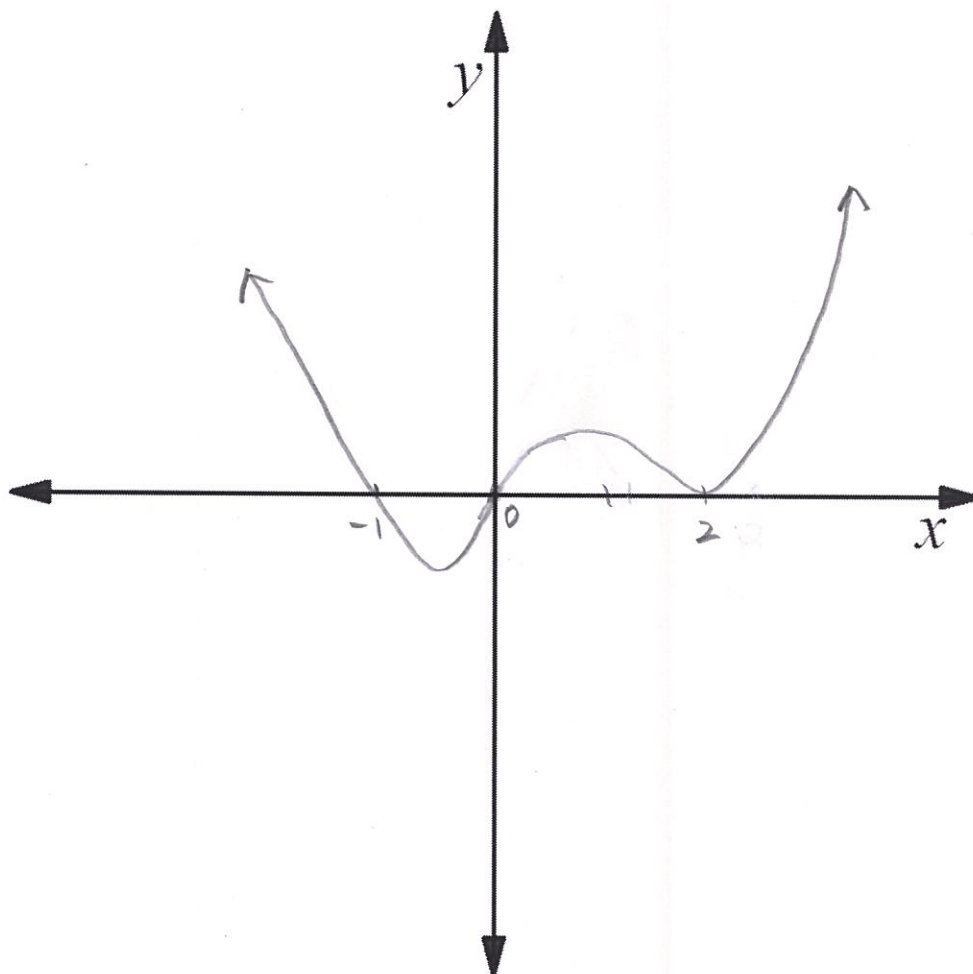
- (c) (i) Show that $x(x+1)(x-2)^2 = x^4 - 3x^3 + 4x$

2

$$\begin{aligned} \text{LHS} &= (x^2+x)(x^2-4x+4) \\ &= x^4 - 4x^3 + 4x^2 + x^3 - 4x^2 + 4x \\ &= x^4 - 3x^3 + 4x \\ &= \text{RHS} \end{aligned}$$

- (ii) Hence, sketch the graph of $P(x) = x^4 - 3x^3 + 4x$

1



- (d) Consider the function $f(x) = \sqrt{9-x^2}$

- (i) State the domain and range of $f(x)$, in interval notation.

2

$$\text{Domain: } [-3, 3]$$

$$\text{Range: } [0, 3]$$

- (ii) State whether the function $f(x)$, is odd, even or neither.
You must justify your answer algebraically.

2

$$f(x) = \sqrt{9-x^2}$$

$$f(-x) = \sqrt{9-(-x)^2}$$

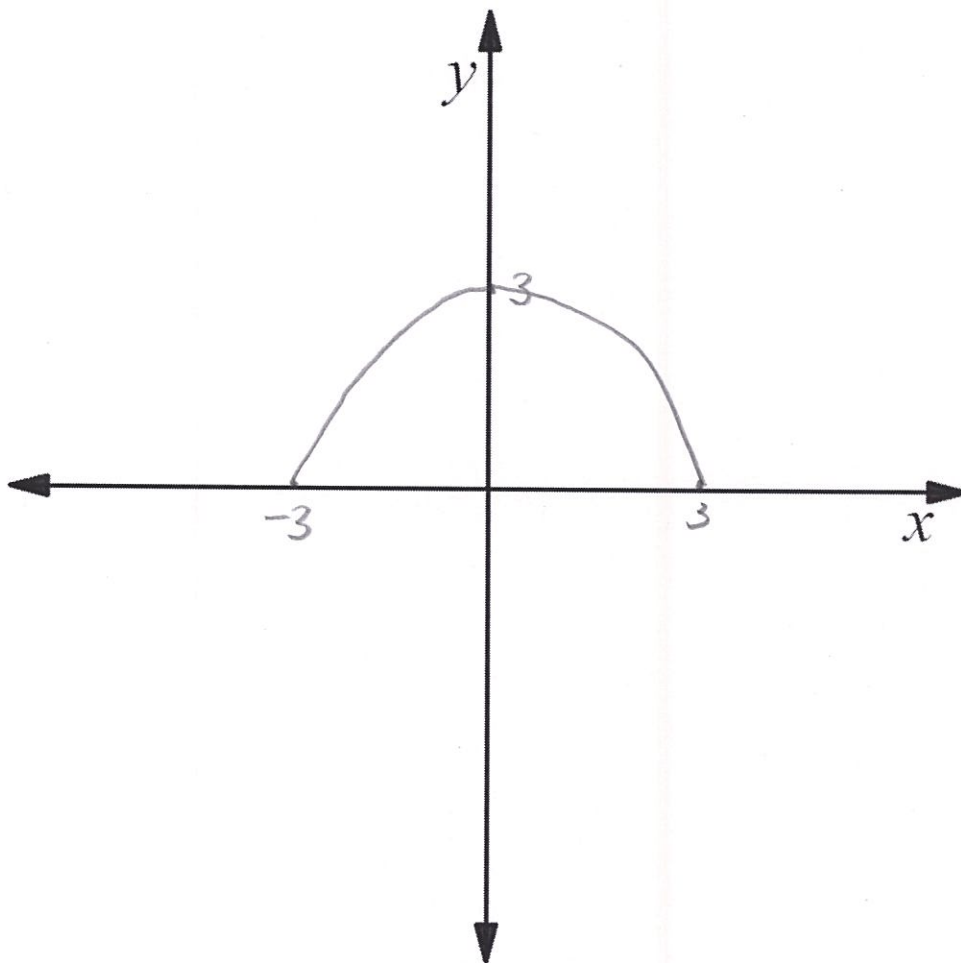
$$= \sqrt{9-x^2}$$

$$= f(x)$$

$\therefore$ even function.

- (iii) Sketch the graph of $f(x) = \sqrt{9-x^2}$, showing all intercepts.

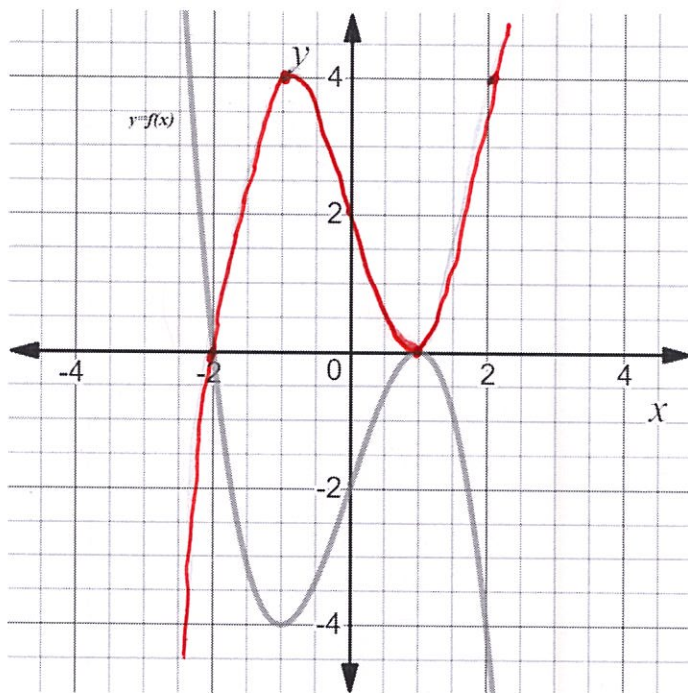
1



(e) The graph of $y = f(x)$ is shown below. Sketch the graph of

(i) $y = -f(x)$ (reflected across x -axis)

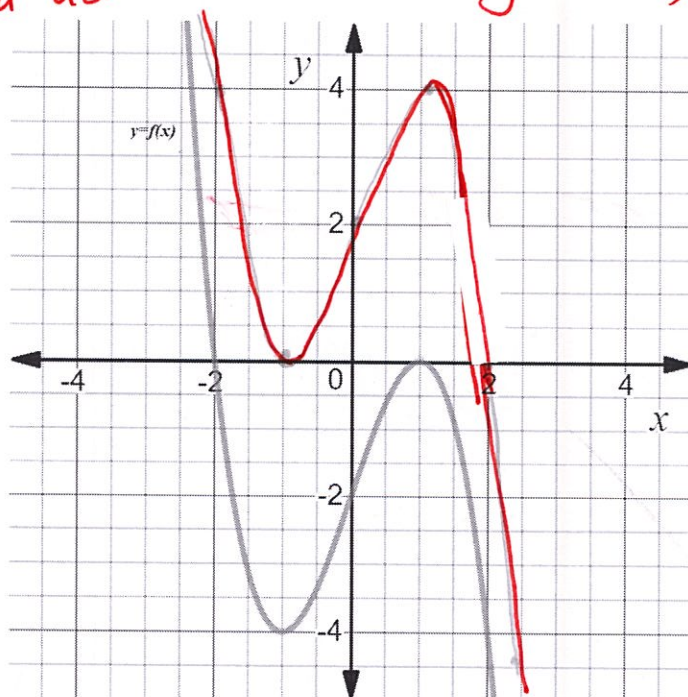
1



(ii) $y = -f(-x)$

(reflected across x -axis + y -axis)

1



End of Paper